

Derivatives of Trigonometric function,

$$1) \frac{d}{dx}(\sin x) = \cos x$$

$$2) \frac{d}{dx}(\cos x) = -\sin x$$

$$3) \frac{d}{dx}(\tan x) = \sec^2 x$$

$$4) \frac{d}{dx}(\operatorname{cosec} x) = -\operatorname{cosec} x \cot x$$

$$5) \frac{d}{dx}(\sec x) = \sec x \tan x$$

$$6) \frac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x$$

Derivatives of Inverse trigonometric function,

$$1) \frac{d}{dx}[\sin^{-1} x] = \frac{1}{\sqrt{1-x^2}}$$

$$2) \frac{d}{dx}[\cos^{-1} x] = -\frac{1}{\sqrt{1-x^2}}$$

$$3) \frac{d}{dx}[\tan^{-1} x] = \frac{1}{1+x^2}$$

$$4) \frac{d}{dx}[\operatorname{cosec}^{-1} x] = -\frac{1}{x\sqrt{x^2-1}}$$

$$5) \frac{d}{dx}[\sec^{-1} x] = \frac{1}{x\sqrt{x^2-1}}$$

Derivatives of Hyperbolic function

$$1) \frac{d}{dx}(\sinh x) = \cosh x$$

$$2) \frac{d}{dx}(\cosh x) = \sinh x$$

$$3) \frac{d}{dx}(\tanh x) = \operatorname{sech}^2 x$$

$$4) \frac{d}{dx}(\operatorname{cosech} x) = -\operatorname{cosech} x \coth x$$

$$5) \frac{d}{dx}(\operatorname{sech} x) = -\operatorname{sech} x \tanh x$$

$$6) \frac{d}{dx}(\coth x) = -\operatorname{cosech}^2 x$$

Derivative of Inverse Hyperbolic function

$$1) \frac{d}{dx}[\sinh^{-1} x] = \frac{1}{\sqrt{1+x^2}}$$

$$2) \frac{d}{dx}[\cosh^{-1} x] = \frac{1}{\sqrt{x^2-1}}$$

$$3) \frac{d}{dx}[\tanh^{-1} x] = \frac{1}{1-x^2}$$

$$4) \frac{d}{dx}[\operatorname{cosech}^{-1} x] = -\frac{1}{x\sqrt{1+x^2}}$$

$$5) \frac{d}{dx}[\operatorname{sech}^{-1} x] = -\frac{1}{x\sqrt{1-x^2}}$$

$$6) \frac{d}{dx}[\coth^{-1} x] = \frac{1}{1-x^2}$$

Integration of Trigonometric function,

$$1) \int \sin x \, dx = -\cos x$$

$$2) \int \cos x \, dx = \sin x$$

$$3) \int \tan x \, dx = \ln(\sec x)$$

$$4) \int \operatorname{cosec} x \, dx = \ln(\csc x - \cot x)$$

$$5) \int \sec x \, dx = \ln(\sec x + \tan x)$$

$$6) \int \cot x \, dx = \ln(\sin x)$$

$$7) \int \operatorname{cosec} x \cot x \, dx = -\operatorname{cosec} x$$

$$8) \int \sec^2 x \, dx = \tan x$$

$$9) \int \csc x \, dx = e^x$$

$$10) \int \sec x \tan x \, dx = \sec x$$

$$11) \int \operatorname{cosec} x \cot x \, dx = -\operatorname{cosec} x$$

$$\rightarrow 1 + \cos 2x = 2 \cos^2 x \quad \rightarrow \cot^2 x = \operatorname{cosec}^2 x - 1$$

$$\rightarrow 1 - \cos 2x = 2 \sin^2 x \quad \rightarrow \tan^2 x = \sec^2 x - 1$$

$$\rightarrow 1 + \cos x = 2 \cos^2 x/2 \quad \rightarrow \cot^2 x = 1 - \operatorname{cosech}^2 x$$

$$\rightarrow 1 - \cos x = 2 \sin^2 x/2 \quad \rightarrow \tan^2 x = 1 - \sec^2 x$$

$$\tanh^2 + \operatorname{sech}^2 = 1$$

$$\cot^2 \theta = \operatorname{cosec}^2 \theta - 1$$

$$\tan^2 \theta = \sec^2 \theta - 1$$

$$y = \sin^{-1} u$$

$$\sin y = u$$

$$\cos y \frac{dy}{du} = 1$$

$$y = \tan^{-1} u$$

$$\tan y = u$$

$$\sec^2 y \frac{dy}{du} = 1$$

$$y = \cos^{-1} u$$

$$\cos y = u$$

$$-\cos y \frac{dy}{du} = 1$$

$$\frac{dy}{du} = \frac{1}{\cos y}$$

$$= \frac{1}{\sqrt{1-\sin^2 y}}$$

$$= \frac{1}{\sqrt{1-u^2}}$$

$$\frac{dy}{du} = \frac{1}{\sec^2 y}$$

$$= \frac{1}{1+\tan^2 y}$$

$$= \frac{1}{1+u^2}$$

$$\frac{dy}{du} = -1$$

$$= -1$$

$$= -\frac{1}{u\sqrt{1-u^2}}$$

$$\cosh u = y$$

$$x = \cosh^{-1} y$$

$$1 = \sinh y \frac{dy}{dx}$$

$$\frac{dy}{dx} = \frac{1}{\sinh y}$$

$$= \frac{1}{\cosh^2 y}$$

$$= \frac{1}{1+\sinh^2 y}$$

$$= \frac{1}{1+x^2}$$

$$\operatorname{sech}^{-1} y = x$$

$$-\operatorname{sech}^{-1} y \frac{dy}{dx} = 1$$

$$\frac{dy}{dx} = -1$$

$$= -\frac{\operatorname{sech}^{-1} y}{1-\operatorname{sech}^2 y}$$

$$= -\frac{\operatorname{sech}^{-1} y}{1-y^2}$$

$$= -\frac{\operatorname{sech}^{-1} y}{1-y^2}$$

$$= -\frac{1}{x\sqrt{1-x^2}}$$

$$y = (\ln u)^n$$

$$\ln y = \ln [\ln(u)^n]$$

$$\ln y = n \ln [\ln(u)]$$

$$\frac{1}{y} \frac{dy}{du} = n \cdot \frac{1}{\ln u} \cdot \frac{1}{u} \frac{du}{du}$$

$$\frac{dy}{du} = \left[\frac{1}{\ln u} + \ln [\ln(u)] \right] \frac{1}{u}$$

$$= \left[\frac{1}{\ln u} + \ln [\ln(u)] \right] \frac{1}{u}$$

$$y = \cosh^{-1} u (\operatorname{sech} u)$$

$$= -1 \cdot \frac{\operatorname{sech} u \tanh u}{\operatorname{sech}^2 u \sqrt{1+\operatorname{sech}^2 u}}$$

$$= -\frac{\tanh u}{\sqrt{1+\operatorname{sech}^2 u}} = -\frac{\operatorname{sech}^2 u}{\sqrt{1+\operatorname{sech}^2 u}}$$

$$= -\frac{\operatorname{sech}^2 u}{\sqrt{1+\operatorname{sech}^2 u}}$$

Derivatives of trigonometric function.

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Derivatives of Inverse trigonometric function.

$$1) \frac{d}{dx}[\sin^{-1} x] = \frac{1}{\sqrt{1-x^2}}$$

$$2) \frac{d}{dx}[\cos^{-1} x] = -\frac{1}{\sqrt{1-x^2}}$$

$$3) \frac{d}{dx}[\tan^{-1} x] = \frac{1}{1+x^2}$$

$$4) \frac{d}{dx}[\operatorname{cosec}^{-1} x] = -\frac{1}{x\sqrt{x^2-1}}$$

$$5) \frac{d}{dx}[\sec^{-1} x] = \frac{1}{x\sqrt{x^2-1}}$$

$$6) \frac{d}{dx}[\cot^{-1} x] = -\frac{1}{1+x^2}$$

Derivatives of Hyperbolic function

$$1) \frac{d}{dx}(\sinh x) = \cosh x$$

$$2) \frac{d}{dx}(\cosh x) = \sinh x$$

$$3) \frac{d}{dx}(\tanh x) = \operatorname{sech}^2 x$$

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$$5) \frac{d}{dx}(\operatorname{sech} x) = -\operatorname{sech} x \tanh x$$

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Derivative of Inverse Hyperbolic function

$$1) \frac{d}{dx}[\sinh^{-1} x] = \frac{1}{\sqrt{1+x^2}}$$

$$2) \frac{d}{dx}[\cosh^{-1} x] = \frac{1}{\sqrt{x^2-1}}$$

$$3) \frac{d}{dx}[\tanh^{-1} x] = \frac{1}{1-x^2}$$

$$4) \frac{d}{dx}[\operatorname{cosech}^{-1} x] = -\frac{1}{x\sqrt{1+x^2}}$$

$$5) \frac{d}{dx}[\operatorname{sech}^{-1} x] = -\frac{1}{x\sqrt{1-x^2}}$$

$$6) \frac{d}{dx}[\coth^{-1} x] = \frac{1}{1-x^2}$$

Integration of Trigonometric function.

$$1) \int \sin x \, dx = -\cos x$$

$$2) \int \cos x \, dx = \sin x$$

$$3) \int \tan x \, dx = \ln(\sec x)$$

$$4) \int \operatorname{cosec} x \, dx = \ln(\operatorname{cosec} x - \cot x) \quad 7) \int \operatorname{cosech} x \, dx = -\coth x$$

$$5) \int \sec x \, dx = \ln(\sec x + \tan x) \quad 8) \int \operatorname{sech}^2 x \, dx = \tanh x$$

$$6) \int \cot x \, dx = \ln(\sin x) \quad 9) \int \operatorname{csch} x \, dx = -\coth x$$

$$10) \int \sec x \tan x \, dx = \sec x \quad 11) \int \operatorname{cosec} x \cot x \, dx = -\operatorname{cosec} x$$

$$\rightarrow 1 + \cos 2x = 2 \cos^2 x \quad \rightarrow \cot^2 x = \operatorname{cosec}^2 x - 1$$

$$\rightarrow 1 - \cos 2x = 2 \sin^2 x \quad \rightarrow \tan^2 x = \sec^2 x - 1$$

$$\rightarrow 1 + \cos x = 2 \cos^2 \frac{x}{2} \quad \rightarrow \cot^2 \frac{x}{2} = 1 - \operatorname{cosech}^2 \frac{x}{2}$$

$$\rightarrow 1 - \cos x = 2 \sin^2 \frac{x}{2} \quad \rightarrow \tan^2 \frac{x}{2} = 1 - \sec^2 \frac{x}{2}$$